Electron Configuration

Configurations of atoms and ions, orbital diagrams and the filling order

Name _____ Class _____ Date _____

USEFUL INFORMATION

Subshell capacities: s 2, p 6, d 10, f 14 Filling order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p

Noble-gas cores: [He] 2, [Ne] 10, [Ar] 18, [Kr] 36

Hund's rule: spread electrons across a subshell's orbitals before pairing.

A. Understanding

1. How many orbitals are there in a p subshell, and how many electrons can the p subshell hold in total?

2. State the Pauli exclusion principle and Hund’s rule in your own words.

B. Atoms

3. Write the full electron configurations of (a) carbon (Z = 6), (b) magnesium (Z = 12) and (c) chlorine (Z = 17).

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4. Write the noble-gas (shorthand) configurations of (a) calcium (Z = 20), (b) iron (Z = 26) and (c) bromine (Z = 35).

5. Identify the elements with these configurations: (a) $1s^2\ 2s^2\ 2p^3$ (b) [Ne] $3s^2\ 3p^5$ (c) [Ar] $3d^{10}\ 4s^2\ 4p^2$

C. Orbital diagrams

6. Draw the orbital diagram (boxes and arrows) for the 2s and 2p subshells of nitrogen. How many unpaired electrons does nitrogen have?

7. How many unpaired electrons are there in (a) a sulfur atom and (b) a neon atom?

D. Ions

8. Write the configurations of (a) O^{2-} (b) Mg^{2+} (c) Fe^{3+}. Which two of these ions have the same configuration?

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E. Challenge

9. The configuration of chromium (Z = 24) is [Ar] $3d^5\ 4s^1$, not [Ar] $3d^4\ 4s^2$. Suggest why, and state how many unpaired electrons a chromium atom has.

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